ELECTROMAGNETIC EFFECTS

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Chapter 1

ELECTROMAGNETIC EFFECTS

TOPIC 3: ELECTROMAGNETIC EFFECTS **12 PERIODS**

Competency: The learner should know and understand how magnetic fields interact with electric fields and the applications of this phenomenon.

LEARNING OUTCOMES The learner should be able to:	SUGGESTED LEARNING ACTIVITIES	SAMPLE ASSESSMENT STRATEGY

a. investigate the behaviour of magnets and magnetic fields (s)
b. understand that a current carrying conductor produces a magnetic field that can be detected. (u, s)
c. understand the application of electromagnets in devices such as motors, bells, and generators (u, s)
d. understand the difference between a.c. and d.c (u)
e. know how a.c. and d.c. can be interconverted using rectifiers and inverters (k)
f. understand the action and applications of transformers (u, s, v/a)
	• In groups, learners revisit prior learning about the characteristics of magnets, the magnetic fields around a bar magnet and electromagnets.
	• In pairs, learners research how to make an electromagnet and use their learning to investigate the relationship between the number of coils and the strength of the magnet, and then make a presentation of their results.
	• In groups, learners research, discuss and explain in a presentation how electromagnet technology is applied in an electric bell, a d.c. motor, a relay, a telephone receiver, and a loudspeaker, using circuits to demonstrate principles where possible.
	• In pairs, learners research the difference between a.c. and d.c. and prepare a presentation to:
	  • explain how a.c. and d.c. can be interconverted
	  • classify domestic appliances according to whether they operate on a.c. or d.c.
	  • explain the advantages of using a.c. in commercial supplies and why a d.c. motor will not work when connected to an a.c. supply
	• In groups, learners research, discuss and design a simple transformer and show how it works using a presentation or poster.
	• In pairs, learners produce a report to explain why transformers are not 100% efficient and how their efficiency can be improved
		• Listen to group and pair discussions about magnetism and pose questions to promote critical thinking.
		• Check understanding through the quality of products: presentation and demonstration.
		• Observe learners involved in activities, offering support to deepen learning and avoid misconceptions.

ICT Support

- The learner can use ICT simulations showing how current is produced in magnetic fields.
- The learner can use an online or downloaded simulation to illustrate how the d.c and a.c motors work.

MAGNETIC EFFECTS OF AN ELECTRIC CURRENT

When current flows in a conductor, it generates a magnetic field around it.
The field pattern obtained can be studied by using iron fillings or plotting compass.

AN EXPERIMENT TO SHOW THE MAGNETIC EFFECT OF AN ELECTRIC CURRENT USING A PLOTTING CAMPUS

A card board is supported horizontally.
A copper wire is passed through the centre of the cardboard.
Iron fillings are sprinkled evenly over the cardboard and current is passed through the wire.
The card is tapped gently.
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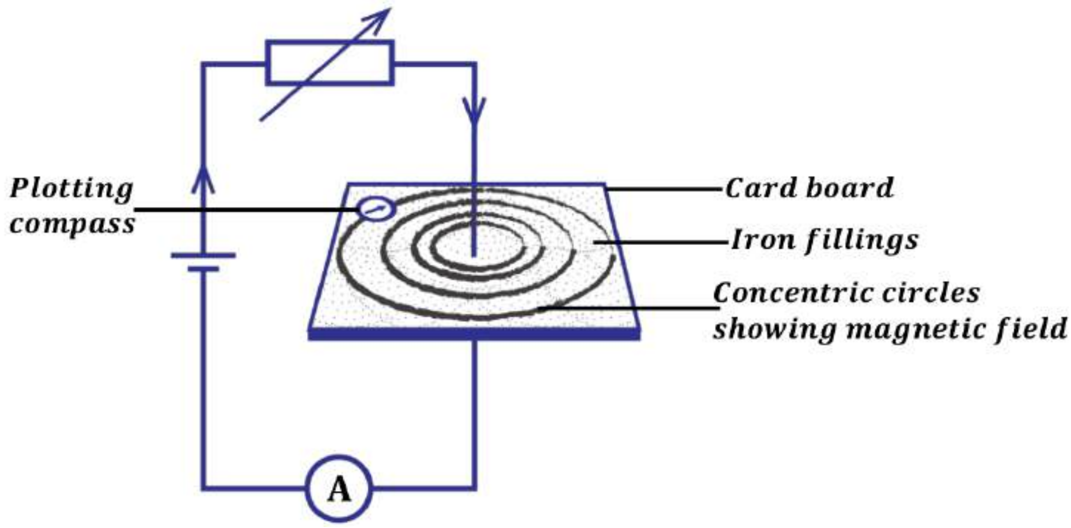
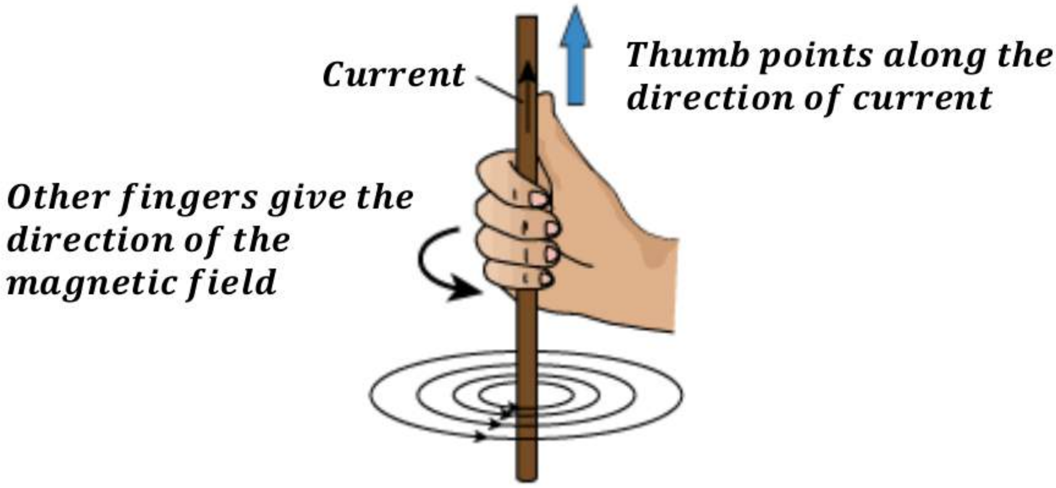
The pattern assumed y the iron fillings is drawn.
A plotting campus is placed at various positions around the wire and the direction of the magnetic field'
The cardboard is again gently tapped and iron fillings settle in concentric circles hence the current carrying conductor has a magnetic field which is concentric.



NB: The direction of magnetic field can be determined by using the **right hand grip rule**

RIGHT HAND GRIP RULE

It states that, " If the wire is grasped in the right hand with the thumb pointing in the direction of current, then the other fingers point in the direction of the mgnetic field.



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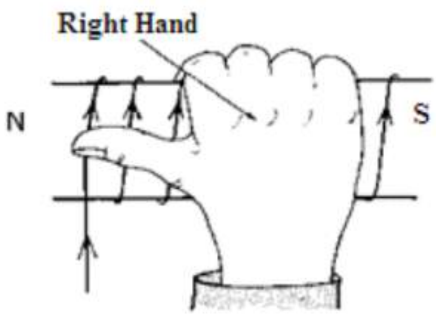
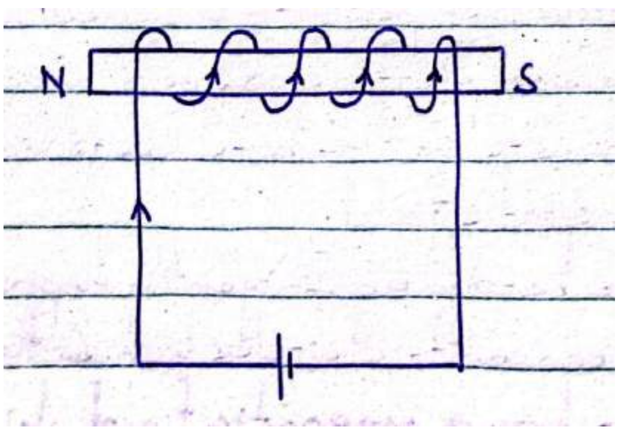
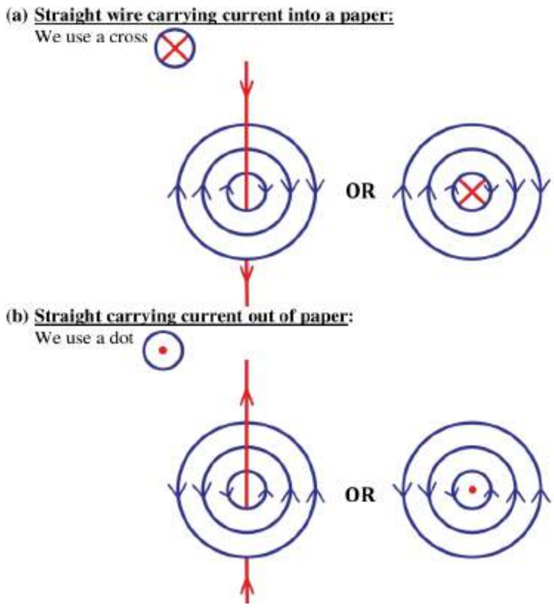
NB: If a solenoid is wound around an iron bar so that the four fingers point in the direction of current, then the thumb points in the north pole.





Magnetic field lines due to a straight wire carrying current



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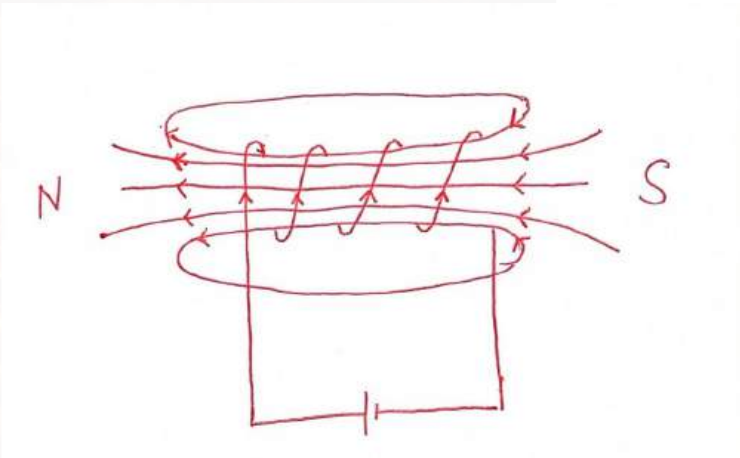
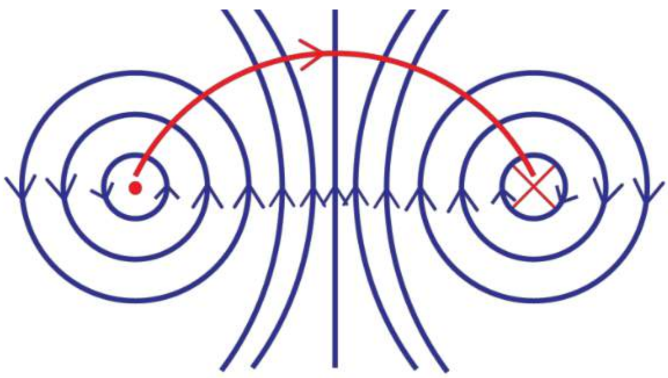
Magnetic field lines due to a solenoid carrying current

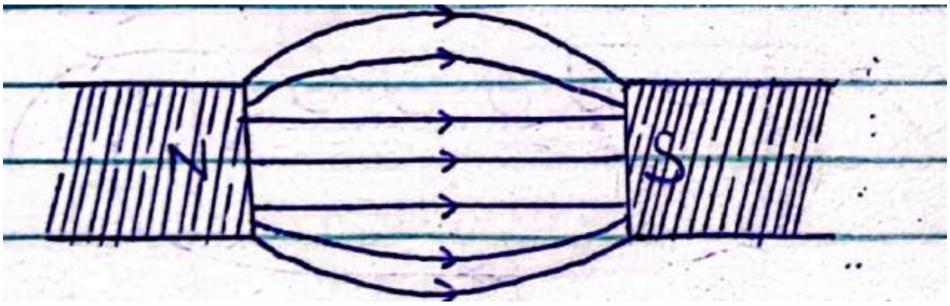


The field pattern due to a solenoid is similar to that of a bar magnet.

Magnetic field lines due to a circular coil carrying current



Magnetic field lines due to a bar magnet



Magnetic field lines due to a wire carrying current

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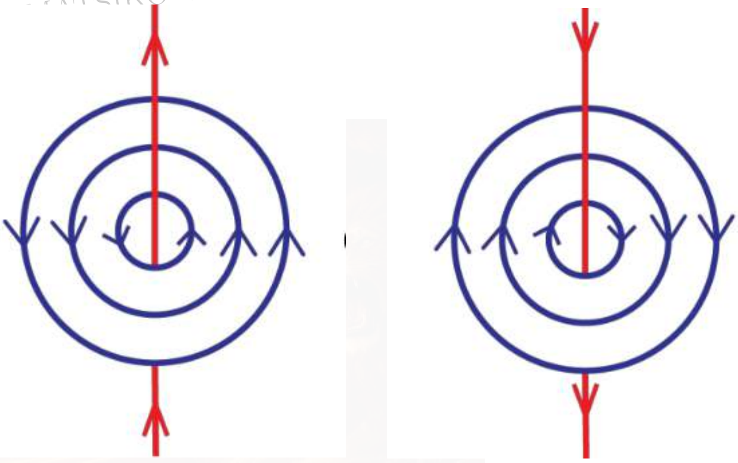
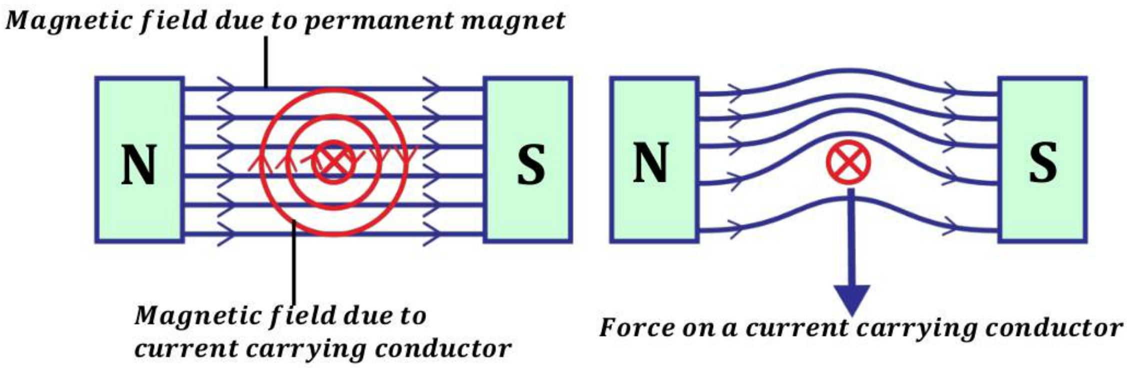


Force on a current carrying conductor in a magnetic field

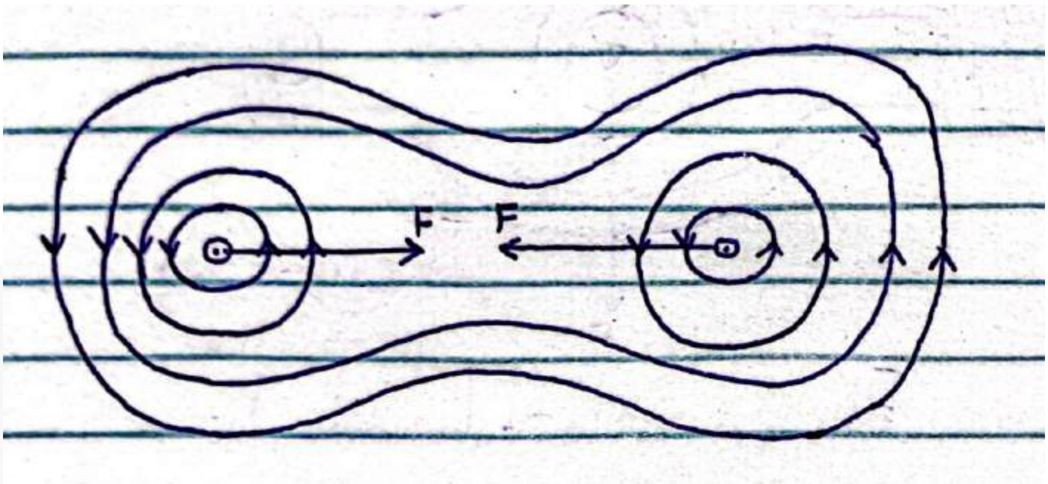


(a) Individual magnetic fields **(b) Combined magnetic fields**

When a conductor carrying current is placed in the magnetic field eg between the poles of the powerful magnet it sets up a magnetic field around itself.
The combined field exerts a force on the wire and this force is towards the region with fewer field lines.

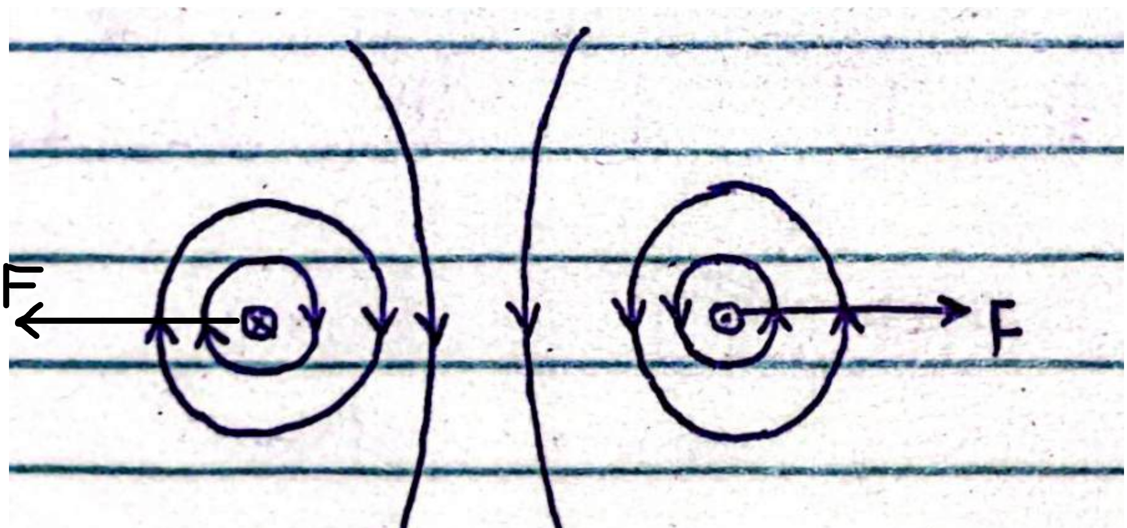
Magnetic field lines due to straight wires carrying current

(a) **Same direction**



The wires attract each other.

(b) **Different directions**



The wires repel each other.

With the currents in the same direction, the field between the wires largely cancels out. There are few field lines to repel hence the wires attract each other.
With currents opposite, the field is greater between the wires and the lines of force repel each other hence the wires repel sideways.

FLEMING'S LEFT HAND RULE

It states that, "If the THUMB, FIRST FINGER and SECOND FINGER are placed at right angles, with the first finger pointing in the direction of the magnetic and second finger in the direction of current, then the thumb points in the direction of force.

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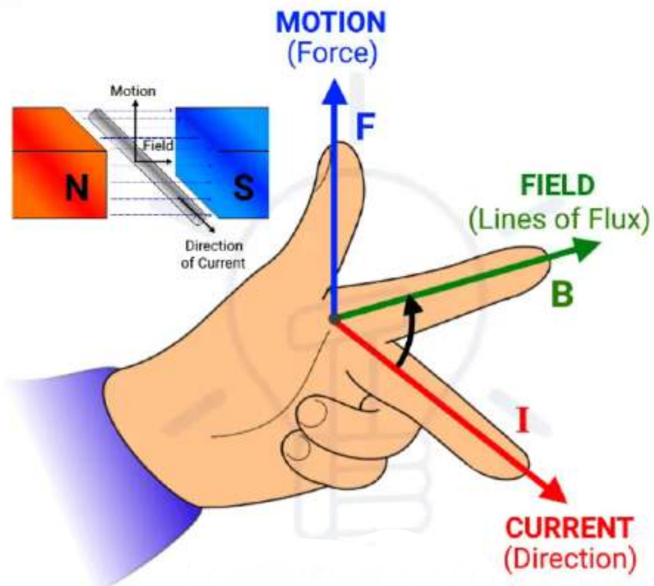
ILLUSTRATION



FACTORS THAT AFFECT THE MAGNITUDE OF FORCE ON A CURRENT CARRYING CONDUCTOR IN A MAGNETIC FIELD

1. **Manitude of current flowing through the conductor.**

The force increases with increase in current i.e

2.**Length of a conductor**

Force increases with increase in length.

3. **Strength of the magnetic field.**

The stronger the magnetic field, the larger the force on a conductor.

4. **Area of the conductor**

The larger the area, the larger the force on a conductor.

APPLICATIONS OF ELECTOMAGNET

1. Lifting magnets.
2. Electric bells.
3. Telephone receiver
4. Magnetic relay switch
5. Moving coil loud speaker
6. Motors
7. Generators

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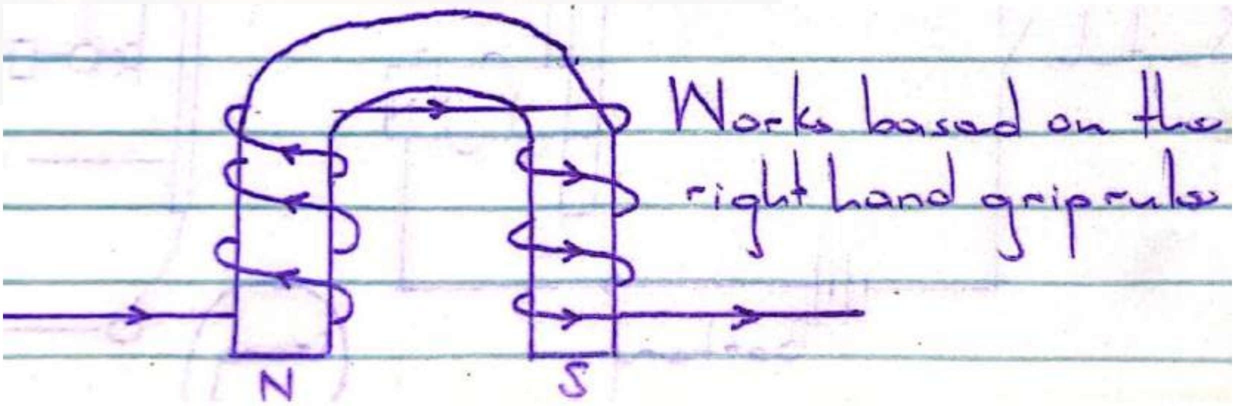
LIFTING MAGNETS

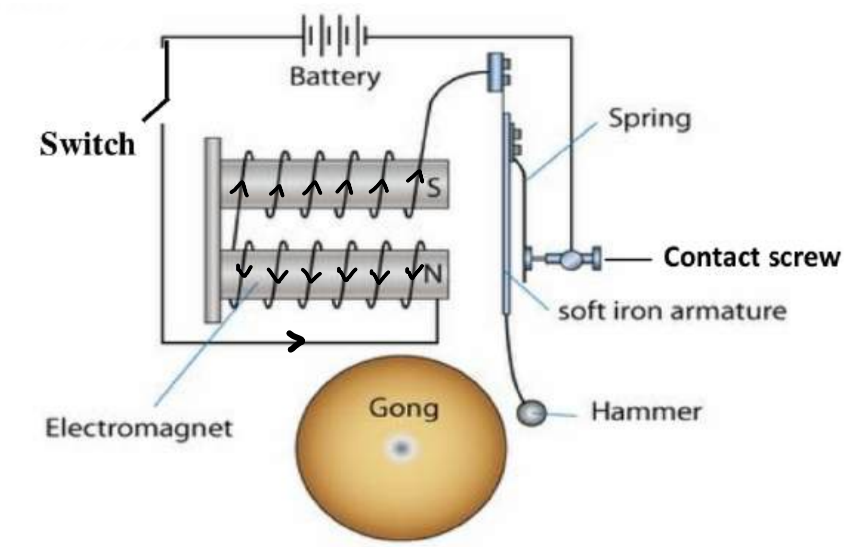
They are used in lifting and transporting heavy steel from one point to another in factories.

The coil made of copper wire is wound on a U-shaped soft iron.

The coil is wound in opposite directions on each side of the soft iron to create opposite poles.

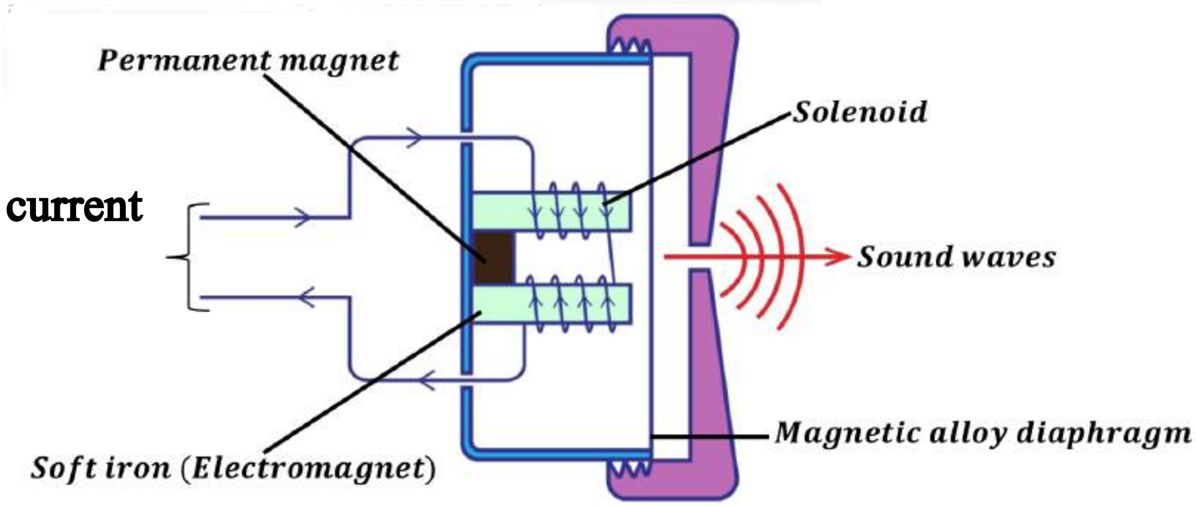


ELECTRIC BELL



When the bell is switched on, current flows in the coil of the electromagnet.
The electromagnet is magnetised and it attracts the armature.
The hammer hits the gong and sound is produced.
As the amarture moves forward, the contact screw is broken.
Current stops flowing and the electromagnet loses its magnetism.
The spring pulls back the amarture to make another contact which completes the circuit again.
This allows the currebt to flow again an the cycle is repeated causing the gong to sound repeatedly.

TELEPHONE RECEIVER (Ear piece)



When a person speaks through a microphone varrying current having the same frequency is set up
The varying current from the microphone passes through a coil of the electromagnet.
The soft iron is magnetised pulls the diaphragm towards it.
The diaphragm vibrates and produces sound waves that are a copy of those tha entered the microphone.

MAGNETIC RELAY SWITCH

This is a switch operated by an electromagnet opened or closed sets of contact.
It uses a small current in a primary circuit to turn on and off a current in the secondary circuit.

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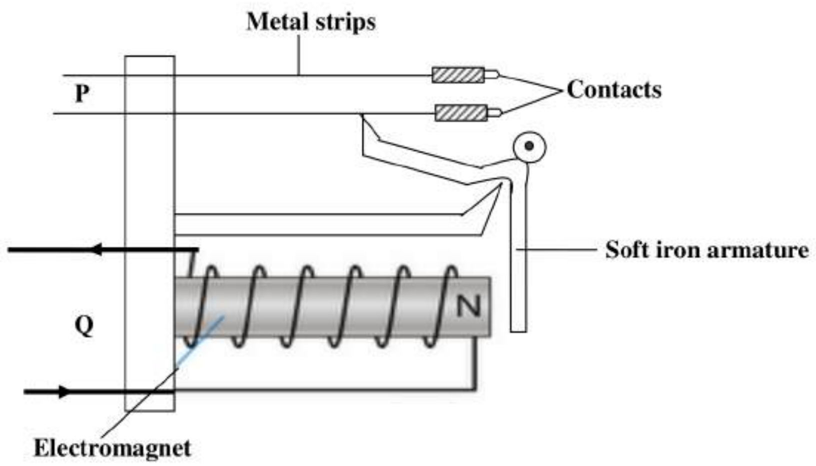
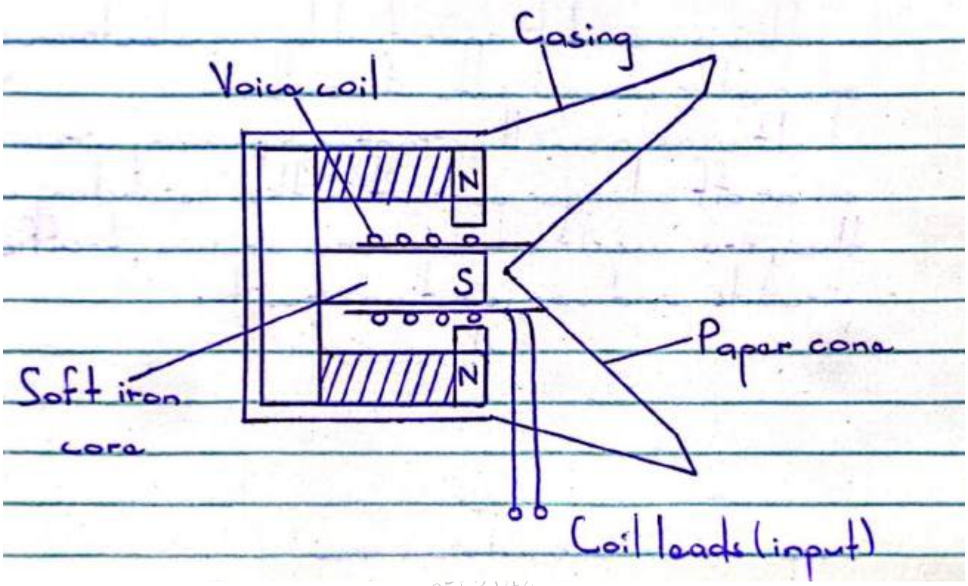


When circuit Q supplies a small current, the electromagnet attracts the soft iron amarture which closes.
This closes the contacts in circuit P by pushing the lower metal strip.
When current in Q is switched off, the electromagnet loses its magnetism and the amarture goes back to its original position hence switching off currrent in circuit P.
NB: They are used in telephone circuits, traffic light circuits and car ignition circuits.

MOVING COIL LOUD SPEAKER



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Varying electric current which corresponds with the sound to be reproduced are passed through the voice coil.

The varying current produces a varying force on the coil according to Fleming's left hand rule.

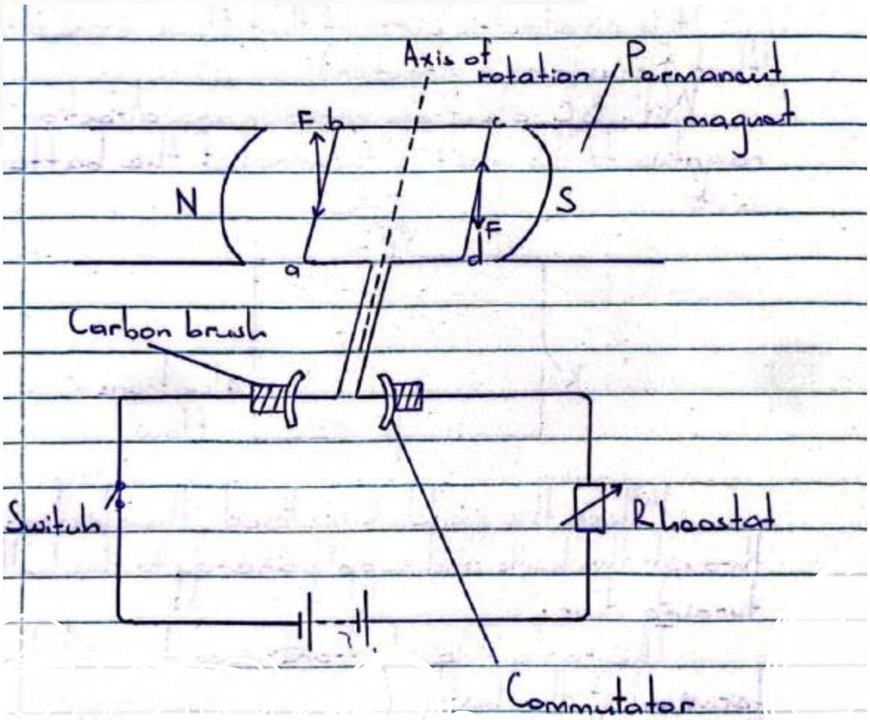
The coil vibrates in and out wit a frequency equal to that of a varying current.

The papercorn attached to the coil vibrates and sets the surrounding air into vibration.

This produces sound.

NB: It is used in radio receivers and radio speakers.

SIMPLE DC MOTOR



When current is switched on, side ab and cd of the coil experience a force, F which is equal in magnitude but opposite in direction.
The two forces form a **couple** which rotates the coil until it is vertical.
In this position, the carbon brushes lose contact with the commutator and current is cut off.
However the coil continues to rotate past this vertical position because of the momentum gained
The current in the coil reverses as the brushes change contact with the commutator, side cd now experiences a downward force and ab an upward force.
Thus the coil continues to rotate as long as the current is flowing.

DIRECT AND ALTERNATING CURRENT

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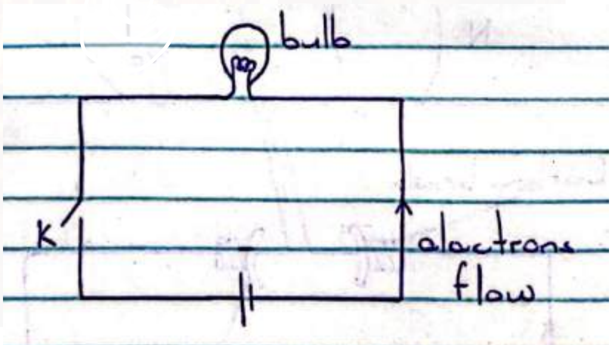
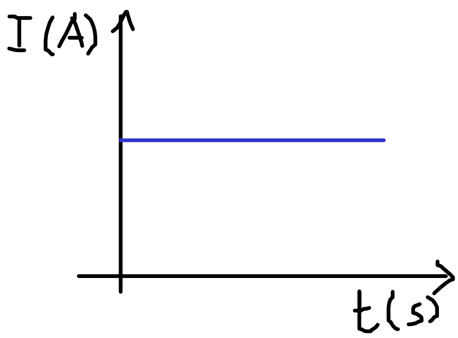
DIRECT CURRENT (DC)

It is an electric current that flows in one direction.
With dc, electrons move in one direction from negative terminal to positive terminal of the battery.
When switck K is closed, electrons flow from through the bulb and supply energy



to the bulb hence turning it on.
The remaining electrons move towards positive terminal of the battery thus electric current flows through a closed circuit.
This produces continues until the battery eventually losses all its charge.
The flow of current is in a constant direction.



SOURCES OF DC

Cells

Batteries

DC generators

Solar cells

Rectified power supplies

ADVANTAGES OF DC

It can be used for low voltage uses eg low voltage lighting in LED lights, charging batteries.

DC can easily be stored in batteries.

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DC provides a steady and stable voltage hence good for sensitive electronic devices eg phones.

Most electronic devices operate with DC thus it works well with circuits.

DISADVANTAGES OF DC

DC power transmission over long distances ca result in high loss.

It requires regular maintainance.

DC power is not suitable for applications that require a constantly changing current.

DC cannot easily be stepped up or stepped down

APPLICATIONS OF DC

Low voltage lighting eg LED lights.

Battery powered devices.

Solar power systems

DC motors

ALTERNATING CURRENT (AC)

With ac, electrons do not really flow. They vibrate from positive to negative terminal and negative to positive.

Charge flows both ways.

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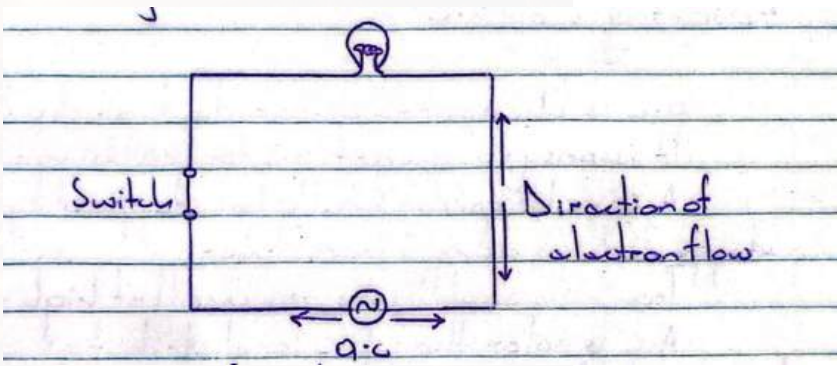
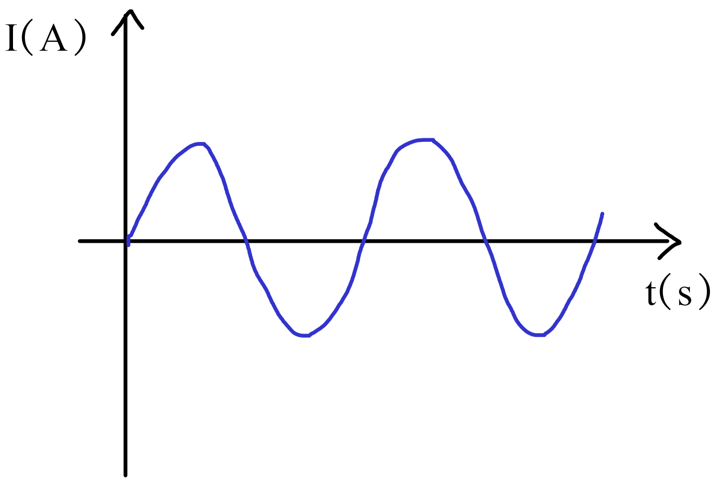
Alternating current (a c) periodically reverses direction and changes its magnitude continuously with time.

When a bulb is connected to ac, it does not have a steady flow of electrons.



Hence light produced by the bulb is not constant BUT this is too fast for the human eye to see.



SOURCES OF AC

AC generators

Power stations

Wind turbines

Inverters

Bicycle dynamos

ADVANTAGES OF AC

AC is cheaper to generate than DC

AC voltage can easily be stepped up or down using a transformer.

It is easy to convert AC to DC using a rectifier.

Transmission losses are small at high voltage.

AC is safer than DC for distribution with a lower risk of electric shock.

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DISADVANTAGES OF AC

Less efficient for low voltage

Not suitable for low voltage and low current applications.

More dangerous in faults and short circuits.

Cannot be stored directly

There is energy loss in conversion to DC

APPLICATIONS OF AC

Power transmission across long distances

RECTIFICATION AND INVERSION (INTER CONVERSION OF AC TO DC)

INVERSION (AC to DC)

It is a process of converting dc power to ac power
The device used to convert dc to ac is called an **inverter**.
An inverter converts dc to ac by switching the direction of dc input back and forth very rapidly.
As a result, dc input becomes ac input.

RECTIFICATION (DC to AC)

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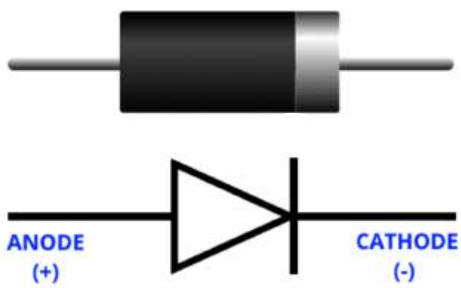
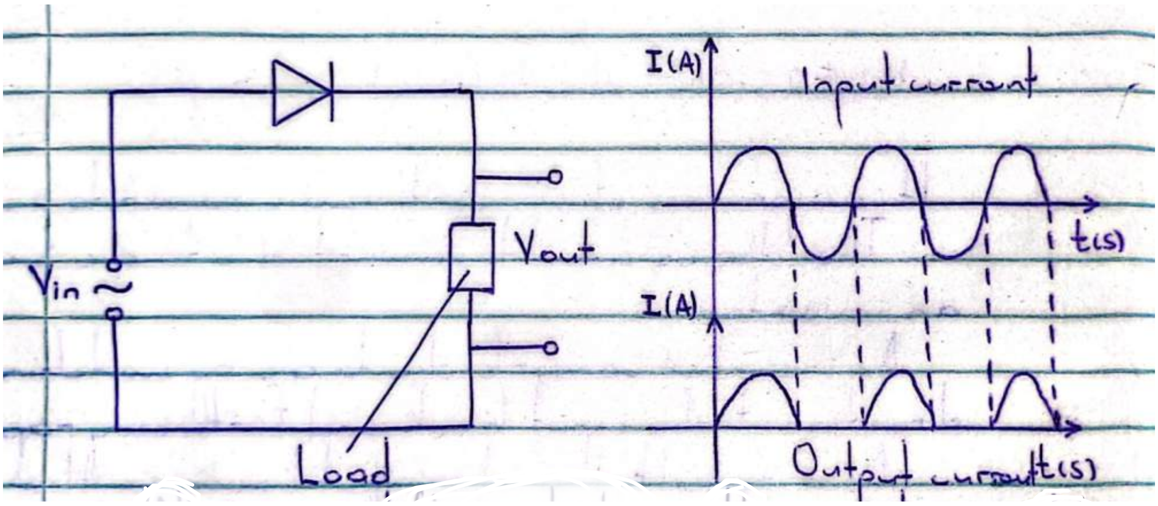
It is the process of converting ac to dc.
It uses diodes to rectify ac voltage allowing current to flow in only one direction and this occurs when the diode is forward biased.
A diode allows current to flow in one direction because it has low resistance in that direction and it is said to be forward biased.
It does not allow current to flow in the other direction because of higher resistance in that direction and it is said to be reverse biased.



TYPES OF RECTIFICATION

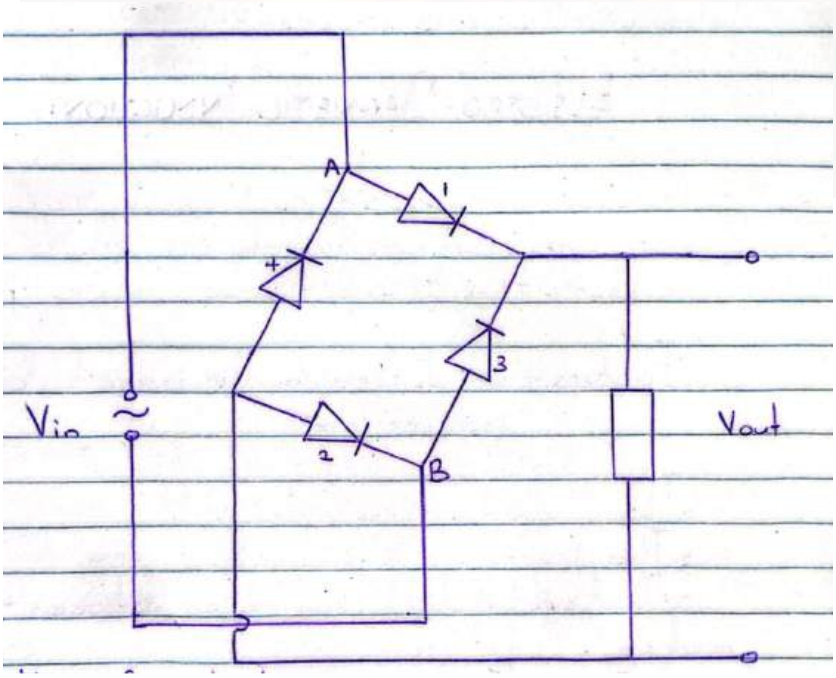
(i) Halfwave rectification
(ii) Fullwave rectification

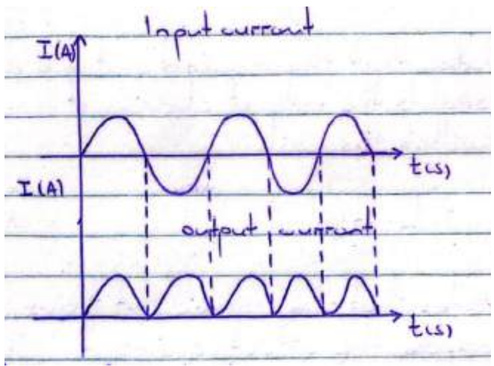
Halfwave rectification



When current flows clockwise, resistance of the diode is low (forward biased) hence current passes through.
When current flows anticlockwise, resistance of the diode is very high (reverse biased) and current is switched off.
This heats of the diode.
AC is converted to DC. Half of the energy is lost and this can be corrected by fullwave rectification.

Fullwave rectification





It uses four (4) diodes. During the first half cycle (clockwise), point A is positive relative to B, current flows throuh diode 1 and 2.
Diode 2 takes back current to the source.
During the second half cycle (anticlockwise), point B is positive relative to A, current flows through diode 3 and 4 takes back current to the source.
Hence current flowing through the load is always in the same direction.

ELECTROMAGNETIC INDUCTION

This is the process by which an electric current is induced in a coil or wire (conuctor) due to change in magnetic flux linking the coil or wire.

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Factors that determine the magnitude of induced e m f

(i) **Number of turns**
The magnitude of induced emf is increased by increasing the number of turns of the coil.
(ii) **Strength of the magnet**
The magnitude of induced emf can be increased by using a strong magnet to provide a stronger magnetic field.
(iii) **Area of coil in magnetic field**
The magnitude of induced emf is increased by placing a large area of coil into the magnetic field.
(iv) **Rate at which magnetic flux changes**
Increasing the speed of motion of a magnet in the coil increases the size of emf induced.

FARADY'S LAW

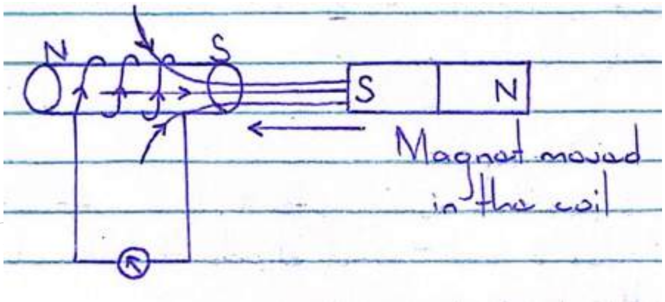
It states that, "The induced emf in a coil is directly proportional to the rate of change of magnetic flux linked within the coil.

LENZ'S LAW

It states that, "The induced emf flows in a direction so as to oppose the effect causing it.

An experiment to prove Lenz's Law

A magnet is moved into the solenoid.
An emf id induced.
The induced current flows in a direction so that the side near the magnet is the south pole hence repulsion.



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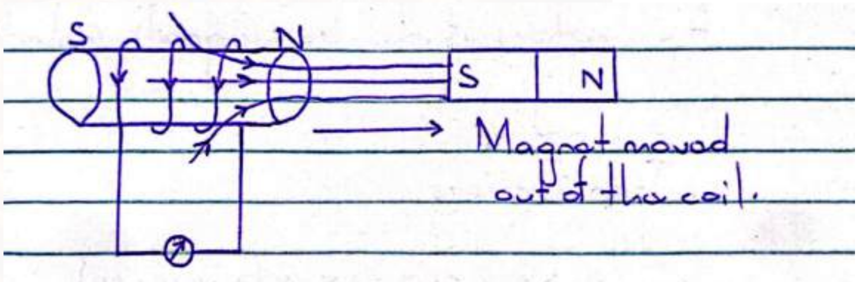
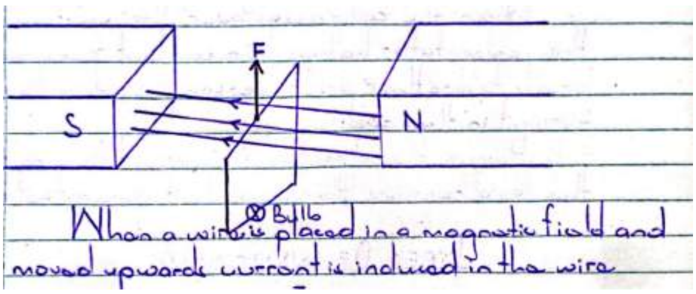
The magnet is then moved out, emf is also induced in the solenoid and the induced current flows in a direction so that the side near the magnet is the North pole hence attraction is experienced.



WIRE MOVED IN A MAGNETIC FIELD

When a wire is placed in a magnetic field and moved upwards, current is induced in the wire the bulb will give light.



Current flows in anticlockwise direction according to Flemings left hand rule.

TYPES OF INDUCTION

1. Self induction
2. Mutual induction

SELF INDUCTION

This is the process by which emf is induced in a coil when changing current current eg AC is passed through the same coil.
The induced emf opposes the emf that induced it.

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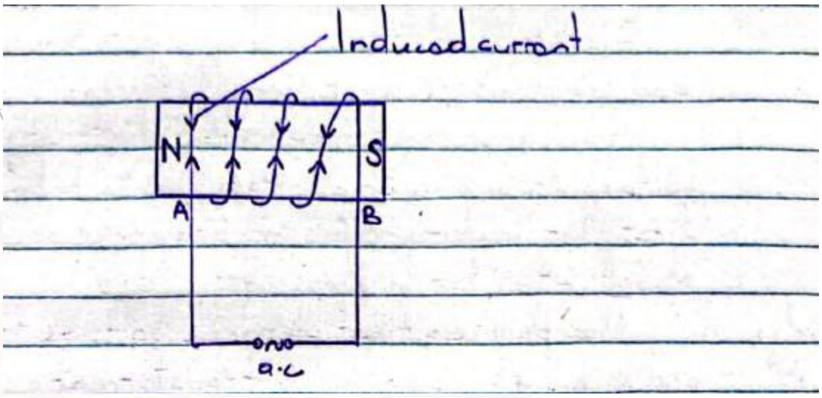
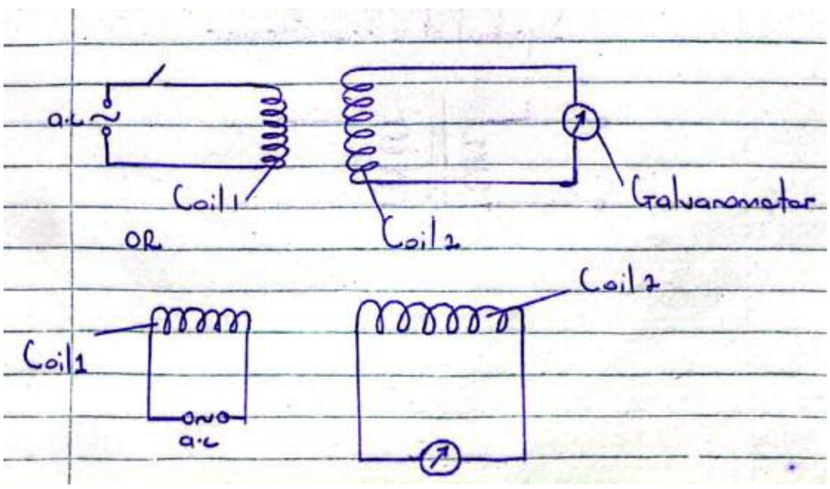


When side A is positve relative to B, the steel bar becomes magnetised so that side A becomes the North pole and side B becomes the South pole.
Due to a change in magnetic flux (caused by AC), a current is induced in the same coil opposing the magnetic field that caused it.

MUTUAL INDUCTION



Mutual induction is the process by which emf is induced in a coil due to changing current in a nearby coil.
In mutual induction, emf is induced in coil 2 (secondary) due to change in current in coil 1 (primar coil).
This is applied in transformers.

APPLICATIONS OF ELECTROMAGNETIC INDUCTION

1. Transformer
2. Generators

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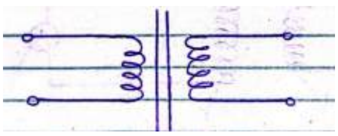
TRANSFORMERS

Electric power is generated at high voltage and transmitted over long distances.
The main power lines carry electricity at a high voltage from power plants which is to be stepped down first for factory use and stepped down further to about 240 V for home use.
A transformer is a device that changes voltage from high voltage to low voltage.

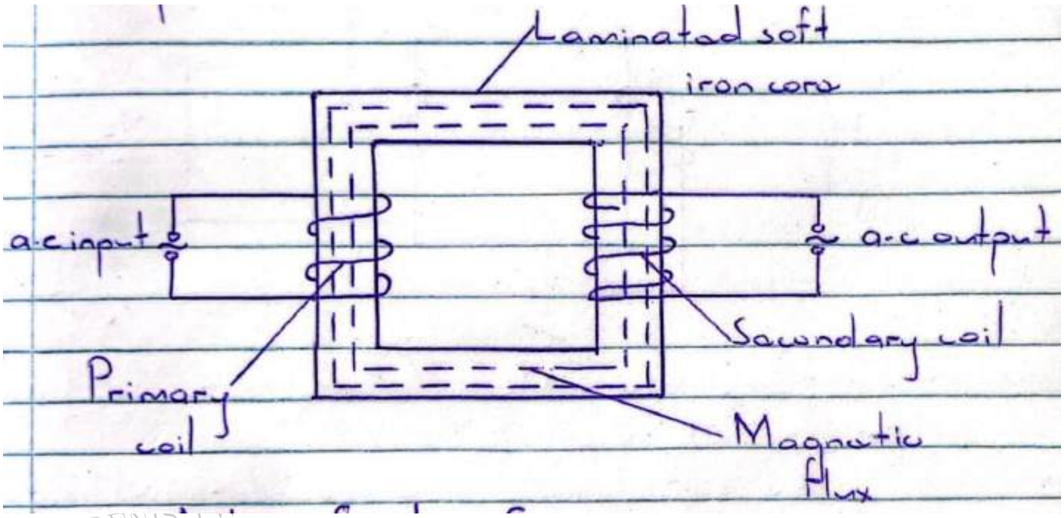
NB: A transformer is used for stepping up and stepping down voltage.
The energy changes in a transformer are: Electrical energy converted to magnetic energy and back to electrical energy.

SYMBOL OF A TRANSFORMER



STRUCTURE OF A TRANSFORMER

A transformer consists of a primary coil and a secondary coil of insulated wire wrapped/ wound on a laminated soft iron core.



ACTION OF A TRANSFORMER

Alternating voltage applied to a primary coil causes an alternating current to flow through it.

The alternating current produces a changing magnetic flux which links with the secondary coil.

An emf (Alternating voltage) is then induced in the secondary coil.

Factors that affect the magnitude of induced emf

(1) Number of turns in the two coils.

(2) Emf applied to the primary coil.

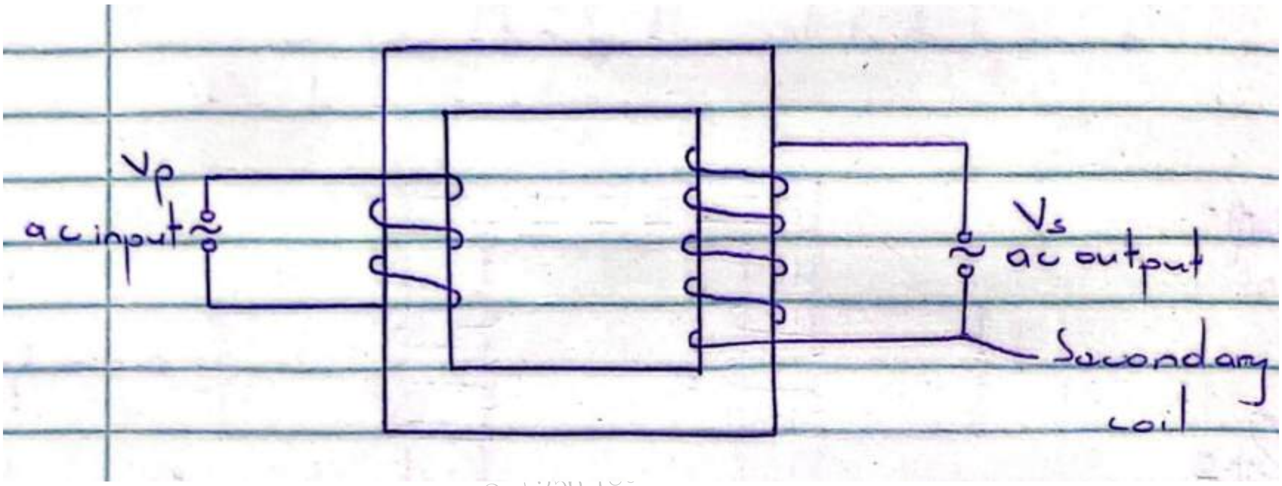
TYPES OF TRANSFORMERS

(1) Step up transformer

(2) Step down transformer.

STEP UP TRANSFORMER

It is a device for increasing alternating voltage.



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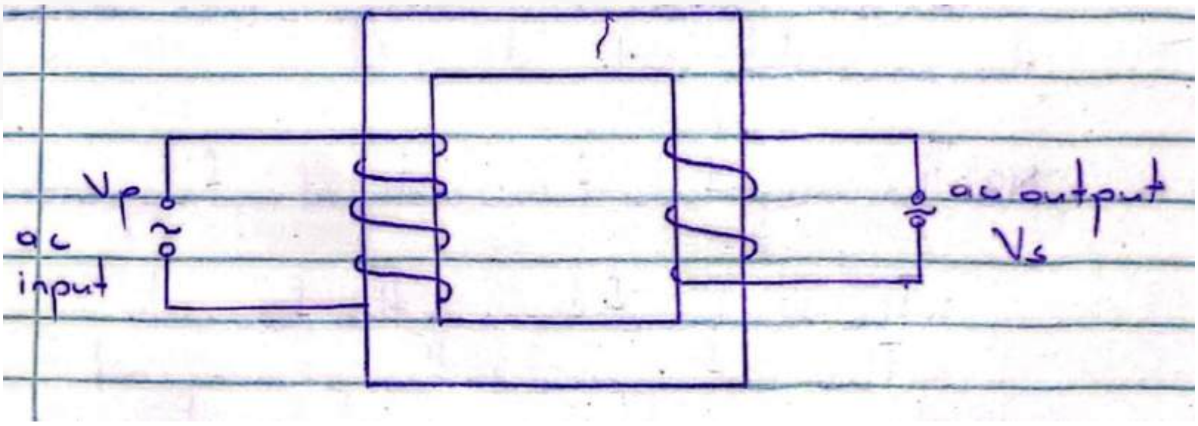
It has more turns in the secondary coil than in the primary coil ie $N_S > N_S$
Emf induced in the secondary coil (output) is greater than emf in the primary coil ie $V_S > V_P$

STEPDOWN TRANSFORMER

It is a device for reducing alternating voltage.



It has more turns in the primary coil than in the secondary coil.

The induced emf in the secondary (output voltage) is less than primary voltage (input voltage).

NB:
(i) A transformer works only with alternating current.
(ii) A transformer works on the principle of electromagnetic induction in the form of mutual induction.
(iii) The magnitude of emf induced in the secondary coil (V_S) can be increased by:

- Increasing the number of turns in the coils.
- Increasing the current in the primary coil.
- Inserting a soft iron core in the coil.

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EFFICIENCY OF A TRANSFORMER

For a transformer which is 100% efficient (an ideal transformer), Power input = Power output. ie ($P_{in} = P_{out}$)
No power is lost.
However, this is not practically possible and there are factors that lead to energy loss in a transformer hence efficiency is always less than one (η ¡ 1)

QN: Explain why transformers are not 100 η efficient?

Power loss in the copper wires due to the resistance of the wounds (coils).
Eddy currents in the transformer core.
The soft iron core is in a changing magnetic field of the primary coil and eddy currents are induced in it which causes heating.
Hysteris losses.
Magnetic flux leakage.

QN: Explain how efficiency can be improved?

Using thick copper wires of low resistance.
Using laminated soft iron core so that current can pass through them.
Using soft iron core.
Wounding the secondary coil on top of the primary coil using an E-shaped core.

TRANSFORMER EQUATIONS

$$\frac{N_p}{N_s} = \frac{V_p}{V_s} \tag{1.1}$$

$$\eta = \frac{P_{out}}{P_{in}} \times 100\% \tag{1.2}$$

NB: For an ideal transformer, $\eta = 100\%$

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$$P_{in} = P_{out} \tag{1.3}$$

$$I_pV_p = I_sI_s \tag{1.4}$$

$$\frac{V_p}{V_s} = N_pN_s \tag{1.5}$$

$$\frac{I_S}{I_p} = N_pN_s \tag{1.6}$$

DENis PHYSICS LABORATORY

Questions

1) What happens when a d.c motor is connected to a.c supply?

solution

It won't rotate at all and it will vibrate as the torque produced by positive and negative cycles cancel out.
Eddy currents are also induced in the core, this causes the motor to heat up.

2) 250 V is applied to a primary coil that has 20 turns, the secondary coil has 1000 turns. What is the voltage in the secondary coil.

3) A 240 V step down transfomer is designed to light 10 lamps rated at 12 V, 20 W each and it blows a current of 1 A in the primary coil. Calculate the efficiency of the transfomer.

4) An a.c transformer of efficiency 80% is connected to a 240 V mains supply. The voltage across the secondary coil of 960 turns is 20 V. Calculate:

(i) number of turns in thr primary coil.

(ii) current in the primary coil if the rsistance of 40Ω is connected across the secondary coil.

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5) A transformer is used on a 240 V a.c supply to deliver 9.0 A at 80 V to a heating coil. If 10% of the energy is dissipated (converted) to heat in the transformer.

(a) What is the current in the primary wounding?

(b) If the number of turns in the primary coil is 100. Find the number of turns in the secondary coil.

6) The power input and power output of a transformer 600 W and 540 W respectively. The transformer has 200 turns in the primary coil and 40 turns in the secondary coil. If the input voltage is 240 V. Calculate

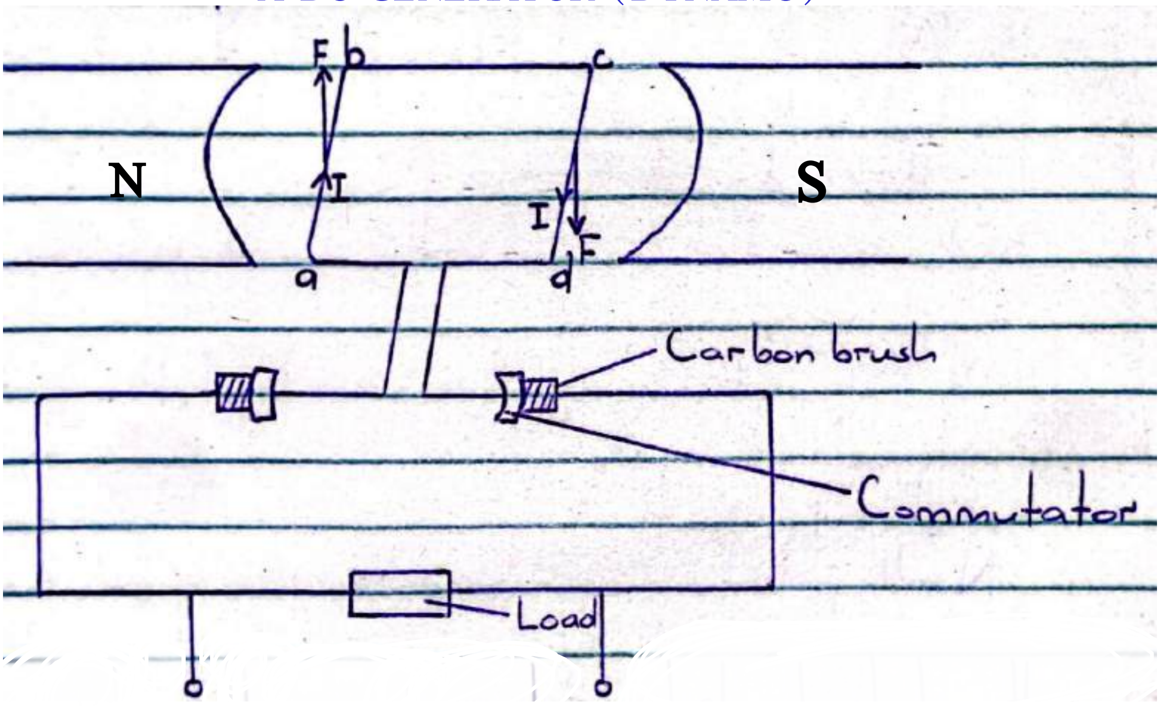
(i) efficiency of the transformer

(ii) current in primary

(iii) current in secondary

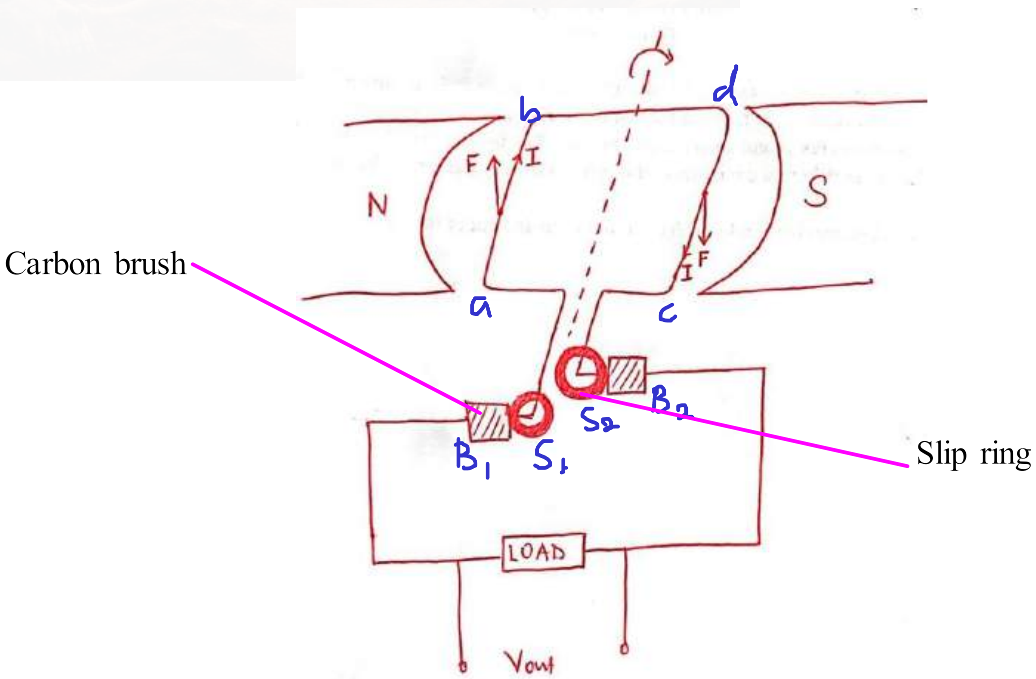
GENERATORS

A DC GENERATOR (DYNAMO)



When the rectangular coil abcd rotates, the magnetic flux linking the coil changes and current flows in the coil. (Fleming's Right rule)
When the coil passes over the vertical position, the commutator halves lose contact from one carbon brush to another and reverses the direction of current in the coil. Hence the direction of current flowing through the load remains the same. (It generates d.c)

AN AC GENERATOR (ALTERNATOR)



The simple a.c generator consists of a rectangular coil, abcd, mounted between, N, S- pole pieces of a strong magnet and freely to rotate with uniform angular velocity.

The ends of the coil are connected to copper slip rings S_1 and S_2, which press against carbon brushes B_1 and B_2.

When the coil rotates with uniform angular velocity in the magnetic field, in accordance to Fleming's Right Hand rule. The magnetic flux density linked with it changes and an emf is induced in the coil. The induced emf is led away by means of the slip rings S_1 and S_2.

When the coil passes over the vertical position, the slip rings change contacts from one carbon brush to another. This reverses the direction of current in the coil. Therefore, the direction of current flowing through the load also changes

END